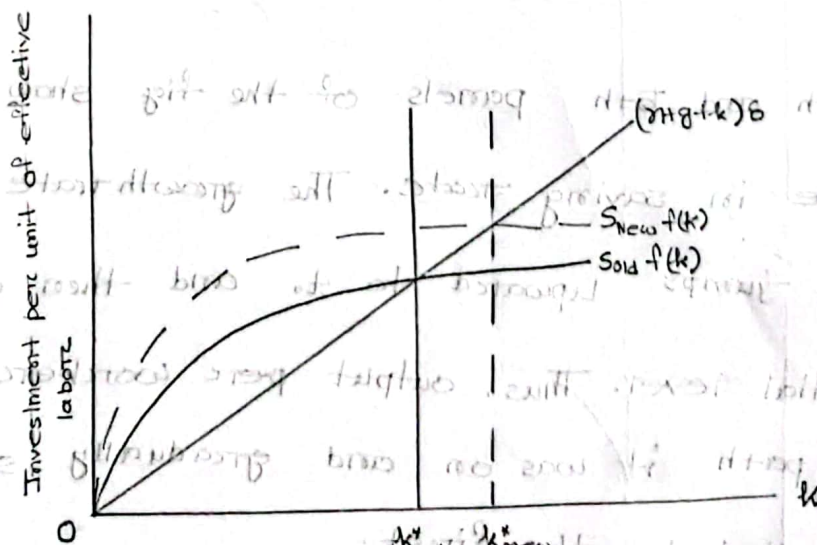


Solow Model

Show the impacts of saving ratio in the steady state solution of solow growth model and prove that exogenous changes in the saving ratio leave only level effects on k and Y but not growth effect.

The solow model focuses on four variables: Output (Y), Capital (k), labor (L) and knowledge or effectiveness of the labor (A). The production function is: $Y(t) = F(k(t), A(t)L(t))$; where t denotes time.

The increase in s shifts the actual investment line upward and so, k^* rises. But k doesn't immediately jump to the new value of k^* . Initially k is equal to the old value of k^* . At this level, actual investment now exceeds breakeven investment - more resources are being devoted to investment than are needed to hold k constant - and so k is positive. Thus k begins to rise. It continues to rise until it reaches the new value of k^* , at which point it remains constant.



These results are summarized in the first three panels of following fig. t_0 denotes the time of the increase in the saving rate. By assumption, s jumps up at time t_0 and remains constant thereafter. Since the jump in s causes the actual investment to exceed breakeven investment by a strictly positive amount, k' jumps from zero to a strictly positive amount. k rises gradually from the old value of k^* to the new value, and k' falls gradually back to zero.

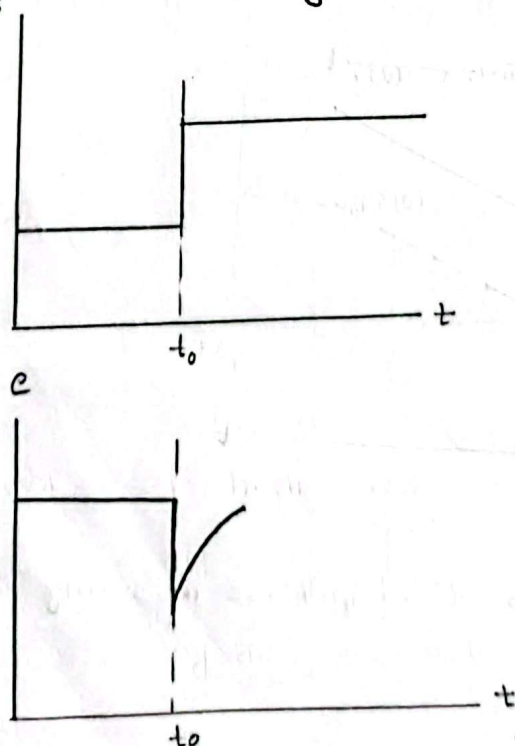
Thus a We know, $Y/L = A f(k)$, when k is constant, Y/L grows at rate g , the growth rate of A . When k is increasing, Y/L grows both because A is increasing and because k is increasing. Thus its growth rate exceeds g . When k reaches the new value of k^* , however, again only the growth of A contributes to growth of Y/L and so growth rate of Y/L returns to g .

The fourth and fifth panels of the fig show how Y/L responds to the rise in saving rate. The growth rate of Y/L which is initially g , jumps up to t and then gradually returns to its initial level. Thus, output per worker begins to rise above its path it was on and gradually settles into a higher path parallel to the first.

In sum, the change in the saving rate has a level effect but not a growth effect. It changes the economy's balanced growth path, and thus the level of output per worker at any point of time, but doesn't affect the growth rate of output per worker on balanced growth path.

✓ Show the impacts of changes in the saving ratio on per capita consumption and longrun output. Does any steady state solution automatically imply golden rule of capital accumulation? Discuss.

Output is divided between investment and consumption and the fraction that is invested is s . Thus consumption per unit of effective labor c equals output per unit of effective labor, $f(k)$, times the fraction of that output that is consumed $1-s$. Since s changes discontinuously at t_0 and k doesn't, initially c jumps downward. It then rises gradually as k rises and s remains at its higher level. This is shown in the following:



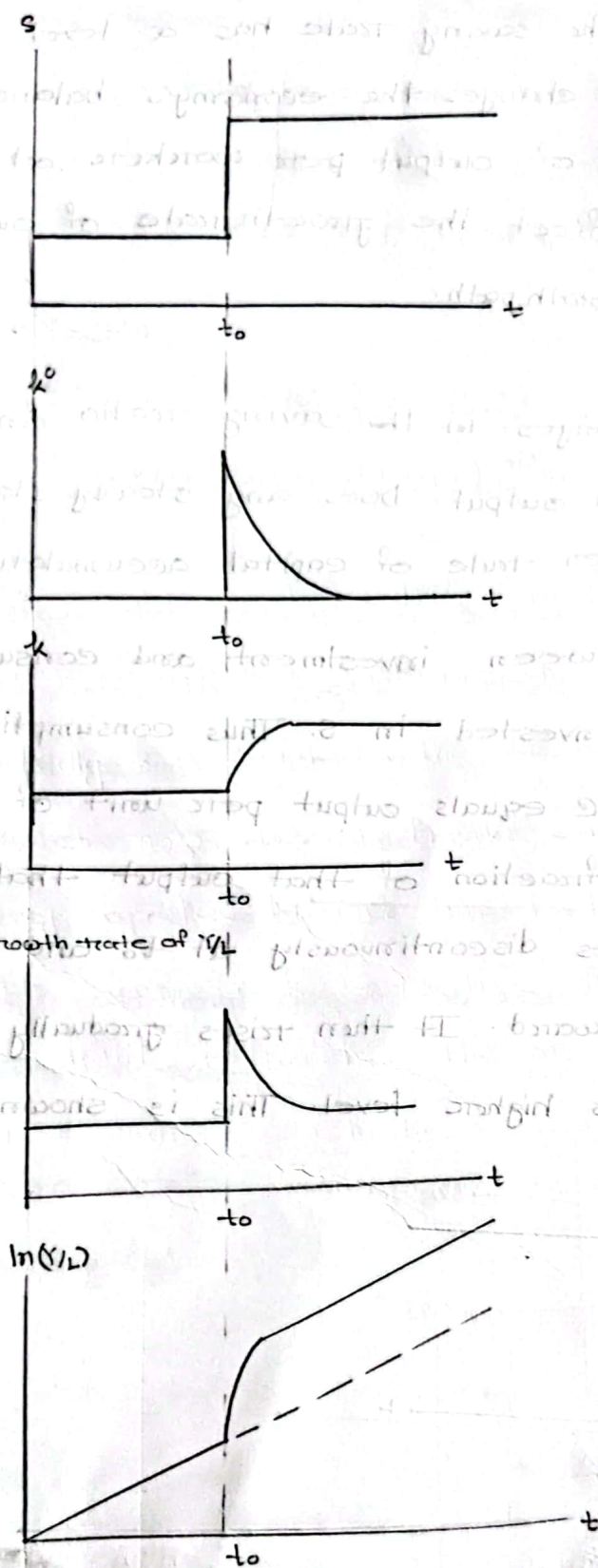


Fig: The effects of an increase in saving rate

Let c^* denote consumption per unit effective labor on balanced growth path. c^* equals output per unit of effective labor $f(k^*)$ minus investment per unit effective labor $sf(k^*)$. On balanced growth path, Actual investment $sf(k^*) = \text{break-even investment } (n+g+s)k^*$.

We know, $Y = c + s$

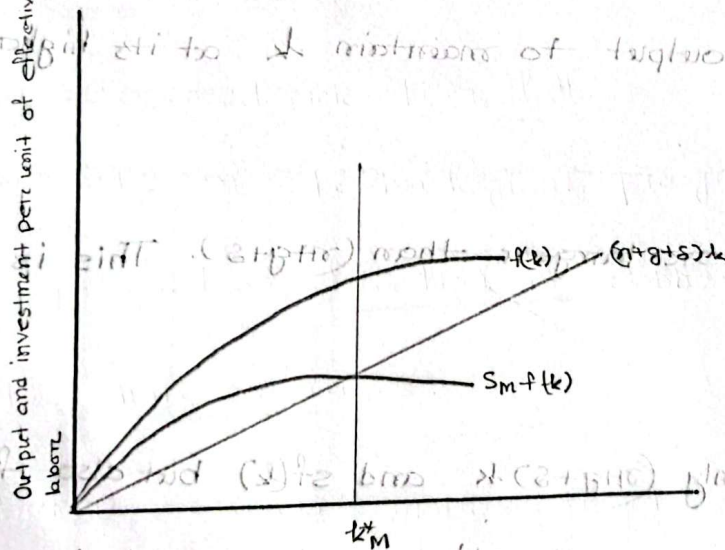
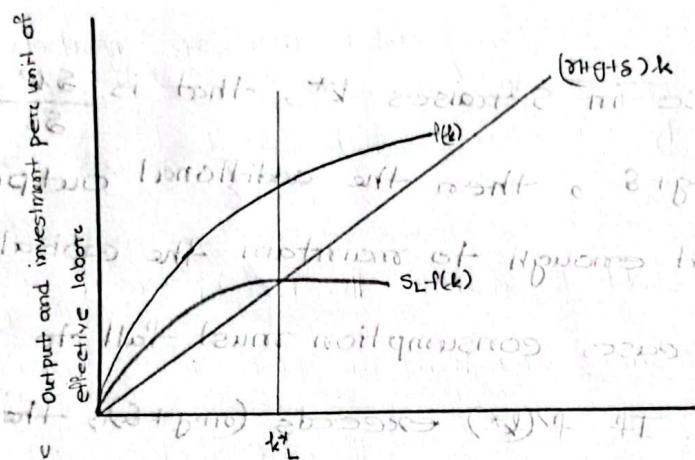
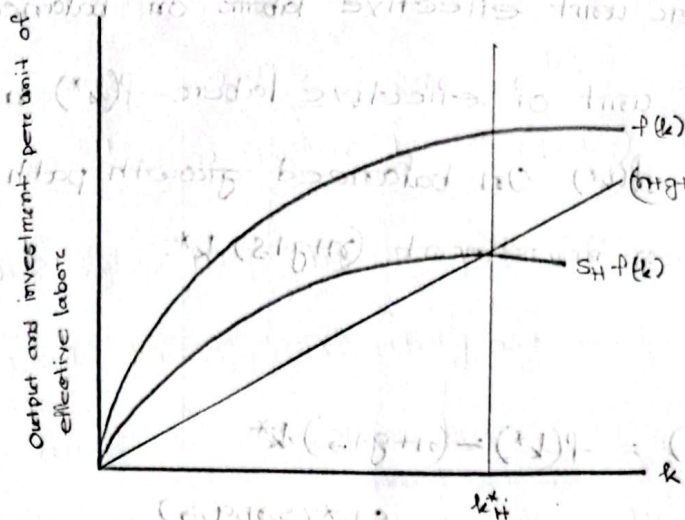
$$\therefore c^* = Y - s = Y - sf(k^*) = f(k^*) - (n+g+s)k^*$$

$$\therefore \frac{\partial c^*}{\partial s} = [f'(k^*)(s, n, g, \delta) - (n+g+s)] \frac{\partial k^*(s, n, g, \delta)}{\partial s}$$

We know that, the increase in s raises k^* , that is $\frac{\partial k^*}{\partial s}$ is positive. If $f'(k^*)$ is less than $n+g+s$, then the additional output from the increased capital is not enough to maintain the capital stock at its higher level. In this case, consumption must fall to maintain the higher capital stock. If $f'(k^*)$ exceeds $(n+g+s)$, there is more than enough additional output to maintain k at its higher level and so, consumption rises.

$f'(k)$ can be smaller or larger than $(n+g+s)$. This is shown in the following fig:

This fig shows not only $(n+g+s)k$ and $sf(k)$ but also $f(k)$. Since consumption on balanced growth path equal output less break-even investment, c^* is the distance between $f(k)$ and $(n+g+s)k$ at $k = k^*$.



In the top panel, s is high and so k^* is high and $f'(k^*)$ is less than $(n+g+\delta)$. As a result, an increase in saving rate lowers consumption even when the economy has reached its new balanced growth path.

In the middle panel, s is low, k^* is low, $f'(k^*)$ is greater than $n+g+s$ and an increase in s raises the consumption in the long run.

Finally, in bottom panel, s is at the level that causes $f'(k^*)$ to just equal $(n+g+s)$ that is $f(k)$ and $(n+g+s)k$ are parallel at $k=k^*$. In this case, a marginal change in s has no effect on consumption in the long run and consumption is at its maximum possible level among balanced growth paths. The value of k^* is known as golden rule of capital.

Find the elasticity of output per unit of effective labor on balanced growth path, y^* , with respect to the rate of population growth, n . If $\alpha_k(k^*) = 1/3$, $g = 2\%$, and $s = 3\%$, by about how much does a fall in n from 2 percent to 1% raise y^* ?

For finding the elasticity of output per unit of effective labor,

- we can use the following equation:

$$\frac{n}{y^*} \frac{dy^*}{dn} = \frac{n}{(n+g+s)} \left[\frac{\alpha_k(k^*)}{1-\alpha_k(k^*)} \right]$$

Now, with $\alpha_k(k^*) = 1/3$, $g = 2\%$, and $s = 3\%$, we need to calculate the effect on y^* of a fall in n from 2% to 1%. Using the midpoint of $n = 0.015$ to calculate the elasticity gives us

$$\frac{\eta}{y^*} \frac{\delta y^*}{\delta \eta} = \frac{0.015}{0.015 + 0.02 + 0.03} \frac{(1/3)}{(1 - 1/3)} \approx -0.12$$

So, this 50% drop in the population growth rate, from 2% to 1%,

will lead to approximately a 6% increase in the level of output

per unit of effective labor, since $(-0.50)(-0.12) = 0.06$. This

calculation illustrates the point that observed differences in

population growth rates across countries are not nearly enough

to account for differences in y that we see.

Describe how, if at all, each of the following developments affect the breakeven and actual investment lines in our basic diagram for the Solow model:

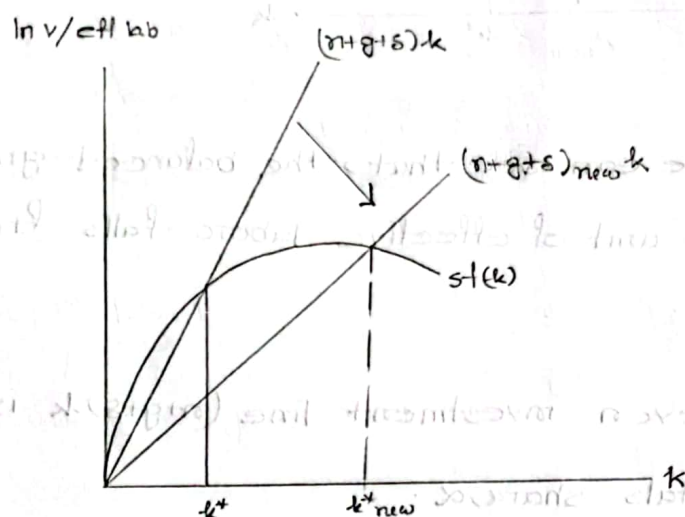
(a) The rate of depreciation falls.

(b) The rate of technological progress rises.

(c) The production function is Cobb-Douglas, $f(k) = k^\alpha$ and capital's share, α rises.

(d) Workers exert more effort, so that output per unit of effective labor is higher than before.

(a) The slope of the break-even investment line is given by $(n+g+s)$ and thus a fall in the rate of depreciation, s , decreases the slope of the break-even investment line. The actual investment curve, $sf(k)$ is unaffected. This is shown in the following fig:

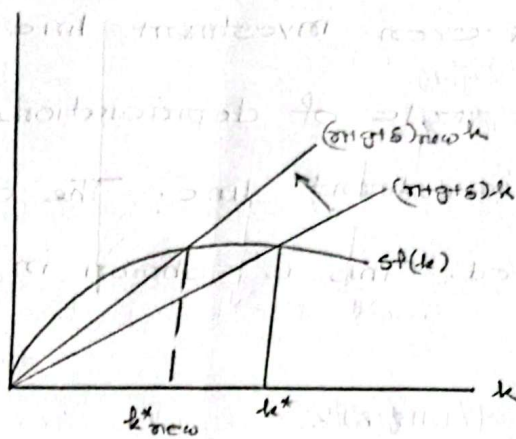


From the fig, we can see that the balanced growth path level of capital per unit of effective labor rises from k^* to k^*_{new} .

(b) Since the slope of the break-even investment line is given by $(n+g+s)$, a rise in the rate of technological progress, g , makes the break-even investment line steeper. The actual investment curve, $sf(k)$ is unaffected.

This is shown by the following fig:

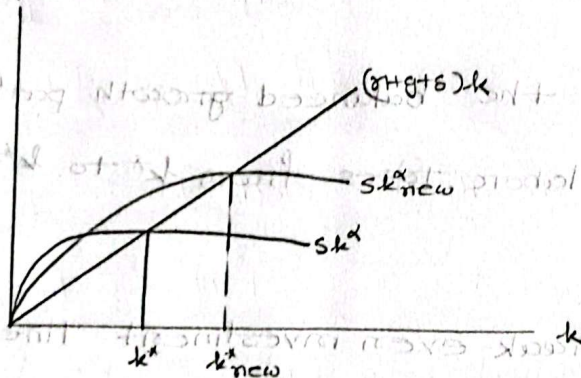
Inv/eff lab



From this fig, we can see that, the balanced-growth path level of capital per unit of effective labor falls from k^* to k^*_{new}

(c) The break-even investment line $(n+g+s)k$ is unaffected by the rise in capital's share, α .

Inv/eff lab



The effect of a change in α on the actual investment curve, $s k^\alpha$, can be determined by examining the derivative $\partial(s k^\alpha)/\partial \alpha$. It is possible to show that

$$\frac{\partial s k^\alpha}{\partial \alpha} = s k^\alpha \ln k$$

For $0 < \alpha < 1$, and for possible values of k , the sign of $\partial(s k^\alpha)/\partial \alpha$

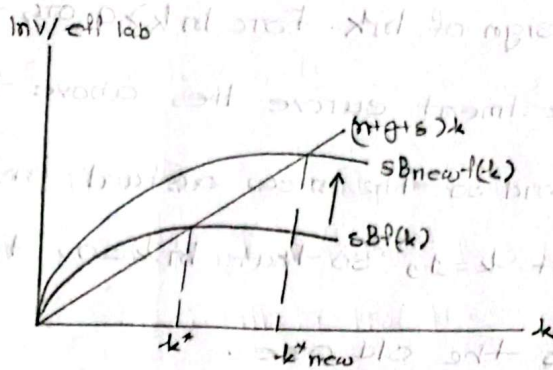
is determined by the sign of $\ln k$. For $\ln k > 0$ or, $k > 1$, $s s k^\alpha / s \alpha > 0$ and so the new actual investment curve lies above the old one. For $\ln k < 0$ or $k < 1$, $s s k^\alpha / s \alpha < 0$, and so the new actual investment curve lies below the old one. At $k=1$, so that $\ln k = 0$, the new actual investment curve intersects the old one.

In addition, the effect of a rise in α on k^* is ambiguous and depends on the relative magnitudes of s and $(n+g+s)$. It is possible to show that a rise in capital's share, α , will cause k^* to rise if $s > (n+g+s)$. This case is depicted in the fig above.

(d) Let us assume that, we modify the intensive form of the production function to include a non-negative constant B , so that actual investment curve is given by $s B f(k)$, $B > 0$.

The workers exerting more effort, so that output per unit of effective labor is higher than before, can be modeled as an increase in B . This increase in B shifts the actual investment curve up. The break-even investment line $(n+g+s)k$, is unaffected.

This is shown in the following fig:



From the fig we can see that the balanced growth-path level of capital per unit of effective labor rises from k^* to k^*_{new} .

Not necessary

Suppose that investment as a fraction of output in the USA rises permanently from 0.15 to 0.18. Assume that capital's share is $1/3$.

(a) By about how much does output eventually rise relative to what it would have been without rise in investment?

(b) By about how much does consumption rise relative to what it would have been without rise in investment?

(c) What is the immediate effect of the rise in investment on consumption? About how long does it take for consumption to return to what it could have been without the rise in investment?

REK

Consider a Ramsey-Cass-Koopmans model with the production function $Y = F(k, AL)$ and real interest rate r , household's utility function takes the form

$$U = \int_0^\infty e^{-\rho t} u(c(t)) \frac{1(t)}{H} dt$$

and there is CRRA (constant-relative-risk-aversion) utility

$$u(c(t)) = \frac{c(t)^{1-\theta}}{1-\theta}$$

(a) Construct household budget constraint and show that there is no-ponzi-game condition.

(b) Solve Euler equation for household's maximization problem.

(c) Describe the behavior of economy in terms of the evolution of c and k .

(d) How does the competitive equilibrium maximize the welfare representative household?

(a) The representative household takes the paths of r and w as given. Its budget constraint is the present value of its life time consumption cannot exceed its initial wealth plus the present value of its lifetime labor income.

Since the household has $L(t)/H$ members, its labor income at t is $w(t) L(t)/H$ and its consumption expenditures are $c(t) L(t)/H$. Its

initial wealth is $1/H$ of total wealth at time 0, or $k(0)/H$. The household's budget constraint is therefore -

$$\int_{t=0}^{\infty} e^{-Rt} c(t) \frac{1}{H} dt \leq \frac{k(0)}{H} + \int_{t=0}^{\infty} e^{-Rt} w(t) \frac{1}{H} dt \dots (i)$$

In general, it is not possible to find the integrals in this expression. Fortunately, we can express the budget constraint in terms of the limiting behavior of the household's capital holdings; and it is usually possible to describe the limiting behavior of the economy. To see how budget constraint can be rewritten in this way can be showed from rearranging eq (i)

$$\frac{k(0)}{H} + \int_{t=0}^{\infty} e^{-Rt} [w(t) - c(t)] \frac{1}{H} dt \geq 0 \dots (ii)$$

We can write the integral from $t=0$ to $t=\infty$ as a limit. Thus eq (ii) is equivalent to

$$\lim_{S \rightarrow \infty} \left[\frac{k(0)}{H} + \int_{t=0}^S e^{-Rt} [w(t) - c(t)] \frac{1}{H} dt \right] \geq 0 \dots (iii)$$

The household's capital holdings at time S are

$$\frac{k(S)}{H} = e^{R(S)} \frac{k(0)}{H} + \int_{t=0}^S e^{R(S)-R(t)} [w(t) - c(t)] \frac{1}{H} dt \dots (iv)$$

Here, $e^{R(S)} \frac{k(0)}{H}$ is the contribution of the household's initial wealth to its wealth at S . The household's saving at t is

$[W(t) - C(t)] \frac{1}{H}$, $e^{-R(s)-R(t)}$ shows how the value of that saving changes from t to s . The expression in (iv) is $R(s)$ times the expression in brackets in (iii). Thus we can rewrite the budget constraint as simply

$$\lim_{s \rightarrow \infty} e^{-R(s)} \frac{K(s)}{H} \geq 0 \quad \dots (v)$$

Expressed in this form, the budget constraint states that the present value of the household's asset holdings cannot be negative in the limit.

Eq (v) is known as the no-ponzi game condition. A ponzi game is a scheme in which someone issues debt and rolls it over forever. That is the issuer always obtains the funds to pay off debts when it comes due by issuing new debt. Such a scheme allows the issuer to have a present value of a lifetime consumption that exceeds the present value of his/her lifetime resources. By imposing eq (i) or (v) we are ruling out such scheme.

(b) The household's problem is to choose the path of $c(t)$ to maximize lifetime utility subject to budget constraint.

We can therefore use the objective function and budget constraint to set up Lagrangian:

$$\mathcal{L} = B \int_{t=0}^{\infty} e^{-\beta t} \frac{c(t)^{1-\theta}}{1-\theta} dt + \lambda \left[K(0) + \int_{t=0}^{\infty} e^{-R(t)} e^{(\pi+g)t} w(t) dt - \int_{t=0}^{\infty} e^{-R(t)} e^{(\pi+g)t} c(t) dt \right]$$

The household chooses c at point in time; that is, it chooses infinitely many $c(t)$'s. The first order condition for individual $c(t)$ is:

$$B e^{-\beta t} c(t)^{-\theta} = \lambda e^{-R(t)} e^{(\pi+g)t}$$

From this we can find:

$$\begin{aligned} \frac{\dot{c}(t)}{c(t)} &= \frac{r(t) - \pi - g - \beta}{\theta} \\ &= \frac{r(t) - \rho - \theta g}{\theta} \dots (i) \end{aligned}$$

The second line uses the definition of β as $\rho - \pi - (1-\theta)g$. To interpret

(i), since $c(t)$ (consumption per worker) equals $\bar{c}(t) A(t)$, the growth rate of c is given by

$$\frac{\dot{c}(t)}{c(t)} = \frac{\dot{A}(t)}{A(t)} + \frac{\dot{\bar{c}}(t)}{\bar{c}(t)}$$

$$= -g + \frac{r(t) - \rho - \theta g}{\theta}$$

$$= \frac{r(t) - \rho}{\theta}$$

Where the second line uses eq(1). This condition states that consumption per worker is rising if the real return exceeds the rate at which the household discount future consumption and is falling if the reverse holds. The smaller is θ - the less marginal utility changes as consumption changes - the larger are the changes in consumption in response to differences between the real interest rate and the discount rate.

eq(i) is known as the Euler equation.

(c) The most convenient way to describe the behavior of the economy is in terms of the evolution of c and k .

For the dynamics of c we can consider the following equation:

$$\frac{\dot{c}(t)}{c(t)} = \frac{r(t) - \rho - \theta g}{\theta} = \frac{f'(k(t)) - \rho - \theta g}{\theta}$$

Thus $\dot{c}$ is zero when $f'(k)$ equals $\rho + \theta g$. Let k^* denote this level of k . When k exceeds k^* , $f'(k)$ is less than $\rho + \theta g$ and so $\dot{c}$ is negative; when k is less than k^* , $\dot{c}$ is positive.

For dynamics of k , in the solow model, $\dot{k}$ equals actual investment minus break-even investment. Since we are assuming that there is no depreciation, break-even investment is $(n+g)k$. Actual investment is output minus consumption, $f(k) - c$. Thus,

$$\dot{k}(t) = f(k(t)) - c(t) - (n+g)k(t)$$

For a given k , $\dot{c}$ is zero when consumption equals the difference between the actual output and breakeven investment line. When c exceeds the level that yields $\dot{c}=0$, k is falling, when c is less than this level, k is rising. The following fig combines these informations:

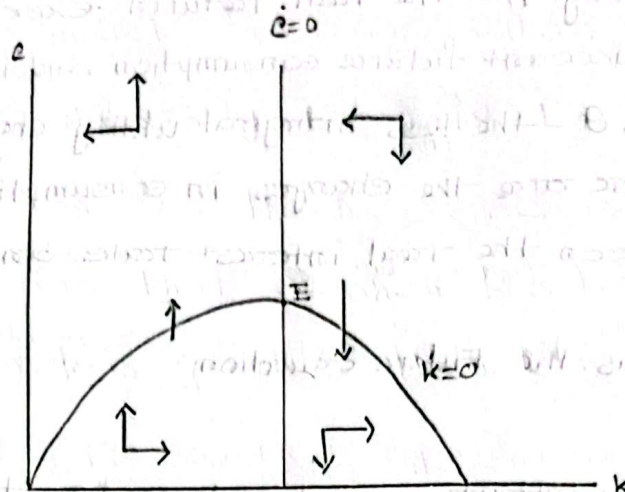


Fig: Dynamics of k and c

The arrows show the direction of motion of both k and c . To the left of the $\dot{c}=0$ locus and above the $\dot{k}=0$ locus, $\dot{c}$ is positive and $\dot{k}$ is negative. Thus c is rising and k is falling. On the $\dot{c}=0$ and $\dot{k}=0$ curves, only one of c and k is changing. On the $\dot{c}=0$ line above the $\dot{k}=0$ locus, c is constant and k is falling. Finally at point E both $\dot{c}$ and $\dot{k}$ are zero, thus there is no movement from this point.

(d) A natural question is whether the equilibrium of the economy represents a desirable outcome. The answer to this question is simple. The first welfare theorem from microeconomics tells us that if markets are competitive and complete and there are

no externalities, then the decentralized equilibrium is Pareto-efficient; that is, it is impossible to make anyone better off without making someone else worse off. Since the conditions of the first welfare theorem hold in the model, the equilibrium must be Pareto efficient.

Since all households have the same utility, this means that the decentralized equilibrium produces the highest possible utility among allocations that treat all households in the same way.

On the contrary, the solution to the social planner's problem is for the initial value of e to be given by the value of the saddle path, and for e and k to then move along the saddle path. That is, competitive equilibrium maximizes the welfare of the representative household.

Compare The speed of convergence between Solow and RCK model.

The speed of convergence in Solow model can be seen by the following equation:

$$\lambda \equiv -\frac{S'k(k^*)}{S k^*} = 1 - \alpha_k(k^*) (\pi + g + s)$$

This means k converges to its balanced growth path at rate $1 - \alpha_k(k^*) (\pi + g + s)$

For the speed of convergence to balance growth path

in RCK we can consider the following eq:

$$u_1 = \frac{1}{2} \left(\beta - \lambda \beta^2 + \frac{1}{\theta} \frac{1-\alpha}{\alpha} (\rho + \theta g) [\rho + \theta g - \alpha(\theta + g)] \right)^{\frac{1}{2}}$$

This adjustment is quite rapid in this case; for comparison, than is solow model for same values of α, n, g . For $\alpha = 1/3, \rho = 4\%, n = 2\%, g = 1\%$ and $\theta = 1$ $u_1 = -5.4\%$, whereas $\lambda = 2\%$. The reason for the difference is that the saving rate is greater than s^* when k is less than k^* and less than s^* when k is greater than k^* . In solow model, in contrast, s is constant by assumption.

RSD

Consider a simplified version of the models of RSD and growth developed by P. Romer (1990), Gross and Helpman (1991) and Aghion and Howitt (1992) where θ reflects the effect of the existing stock of knowledge on the success of RSD.

- (a) Explain the simplification process to construct the framework.
- (b) Critically explain the role of a_L (labour force used in RSD) for the dynamics of knowledge accumulation without considering capital into the model.
- (c) Assume that $\theta + \beta < 1$ and $n > 0$, and that the economy is on its balanced growth path. Describe how each of the following changes affects the $g_A = 0$ and $g_K = 0$ lines and the position in the economy in (g_A, g_K) space at the moment of change:
- (i) an increase in n
 - (ii) " " " α_K
 - (iii) " " " θ
- (d) Assume that $\beta + \theta = 1$ and $n = 0$,
- i) Find the value that A/k must have for g_K and g_A to be equal
 - ii) Find the growth rate of A and k when $g_K = g_A$
 - iii) How does an increase in s affect the long run growth rate of the economy?

(a) The economy has two sectors: good-producing sector and research and development (or R&D) (knowledge-producing) sector. The model uses a standard production function combining labor, capital and technology to produce technological improvements, assuming more resources in research yield more discoveries. Randomness is ignored for long-term growth analysis, and a shift parameter can adjust the success of R&D without explaining what influences research success.

The model simplifies further by using generalized Cobb-Douglas functions for production and may or may not have CRTS. Following the Solow model, it assumes constant, exogenous fractions of output saved and labor/capital used in R&D.

We assume a Cobb-Douglas CRTS production function for goods:

$$Y(t) = [(1-\alpha_k) K(t)]^\alpha [A(t) (1-\alpha_L) L(t)]^{1-\alpha} \quad 0 < \alpha < 1$$

Knowledge is produced according to a Cobb-Douglas that may or may not have CRTS:

$$A(t) = B [\alpha_k k(t)]^\beta [\alpha_L l(t)]^\gamma A(t)^\theta, \quad B > 0, \beta \geq 0, \gamma \geq 0$$

Here B is a shift parameter. θ represents the effect of existing stock of knowledge on R&D success. If past discoveries aid future ones, $\theta > 0$. If easier discoveries are made first, making new ones harder, $\theta < 0$.

(b) To eliminate capital from the model we have to consider the following equation:

$$Y(t) = A(t) (1-\alpha_L) L(t) \dots (i)$$

Similarly the production function for new knowledge is

$$\dot{A}(t) = B [a_L L(t)]^\gamma A(t)^\theta \dots (ii)$$

Population growth Eq (i) implies that output per worker is proportional to A and the growth rate of output per worker equals growth rate of A . The growth rate of A , denoted g_A is:

$$g_A(t) \equiv \frac{\dot{A}(t)}{A(t)} = B a_L^\gamma L(t)^\gamma A(t)^{\theta-1} \dots (iii)$$

From this
$$\frac{\dot{g}_A(t)}{g_A(t)} = \gamma n + (\theta-1)g_A(t)$$

$$\Rightarrow \dot{g}_A(t) = \gamma n g_A(t) + (\theta-1)[g_A(t)]^2 \dots (2)$$

The initial values of L and A and the parameters of the model determine the initial value of g_A . To describe further how the growth rate of A behaves, we must distinguish among the cases $\theta < 1$, $\theta > 1$ and $\theta = 1$.

$\theta < 1$

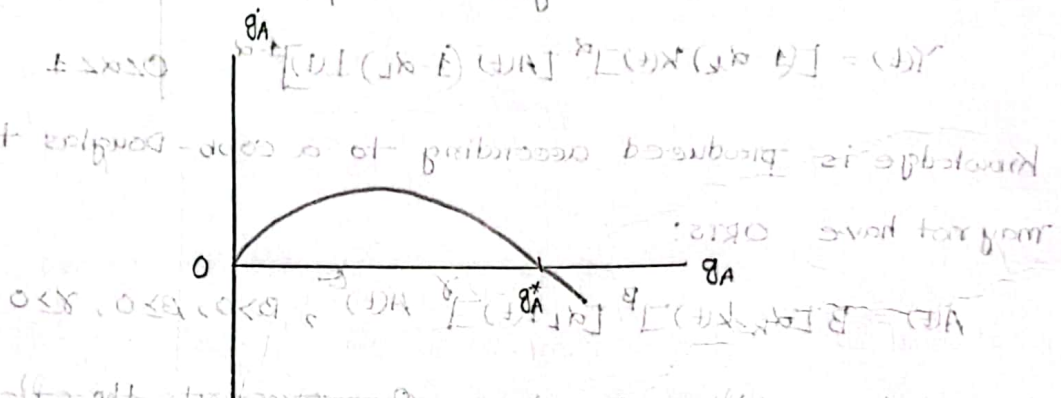


Fig 1: The dynamics of the g_A when $\theta < 1$

Fig-1 shows the phase diagram for g_A when $\theta < 1$. It plots $\dot{g}_A$ as a function of A . The production function for knowledge implies that g_A is always positive. So, the fig considers only positive value of g_A . In the fig considers only positive value of g_A . $\dot{g}_A > 0$ for $g_A > 0$ and small; $\dot{g}_A < 0$ for $g_A > 0$ and large. The unique positive value of g_A where $\dot{g}_A = 0$ is denoted by g_A^* .

where $g_A^* = \frac{\gamma}{1-\alpha}n$

The longrun growth rate of output per worker (g_A^*) increases with the population growth rate n , requiring positive growth for Y/L growth, However countries with faster population growth do not always have higher Y/L growth.

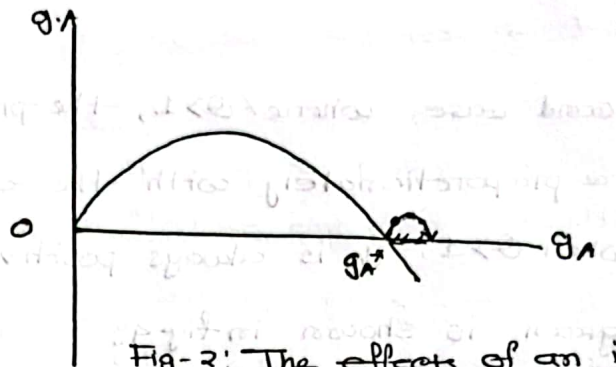
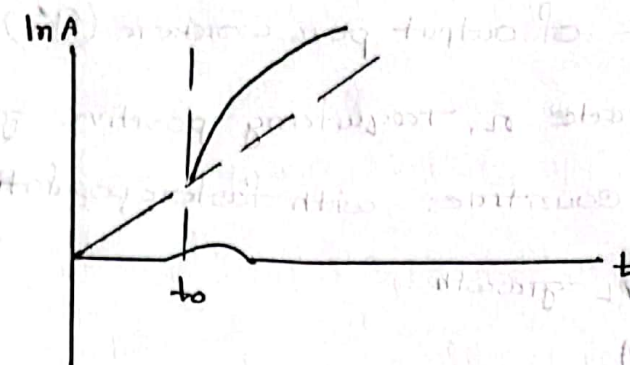


Fig-2: The effects of an increase in α_L when $\alpha < 1$

The above equation indicates that the fraction of labor force in R&D does not affect long-run growth, which is counterintuitive. Suppose that there is a permanent increase in α_L starting from the situation where $g_A = g_A^*$. This change is analyzed in fig 2.

In eq (3) there is no α_L . Thus the rise in α_L doesn't affect the longrun curve showing g_A^* as a function of g_A . But we see α_L in eq (iii), so, increase in α_L causes an immediate increase in g_A . This is shown by the dotted arrow in fig-2. Moreover the growth rate of knowledge is not sustained. When g_A is above g_A^* , g_A is negative, causing g_A to return to g_A^* and then remains there. This is shown by the solid arrows in fig-2. This behavior means that increase in α_L has a temporary impact on g_A , similar to the effect

of a rise in saving rate in the Solow model. This means it has a level effect but not a growth effect. This is shown in fig-3



$\theta > 1$ In second case, where $\theta > 1$, the production of new knowledge increases more proportionately with the existing stock of knowledge. From eq (3), when $\theta > 1$, $\dot{g}_A$ is always positive and increase with g_A .

The phase diagram is shown in fig-4:

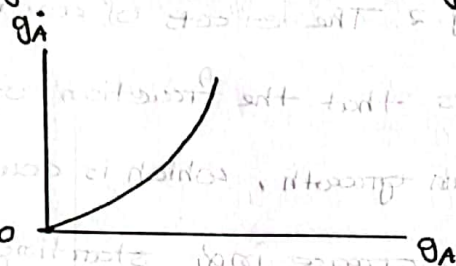


Fig-4: The dynamic of g_A when $\theta > 1$

In this scenario, the economy experiences ever increasing growth instead of converging to a BGP. This is because, knowledge becomes increasingly effective in producing new knowledge, causing the g_A to continually rise. Once knowledge accumulation starts, the economy enters a path of ever increasing growth.

An increase in the fraction of the labor force in R&D, α_L , has a significant impact. From eq (ii), an increase in α_L immediately raises g_A . Since $\dot{g}_A$ is an increasing function of g_A , $\dot{g}_A$ also rises in α_L . Results in a continuously increasing g_A .

$\theta = 1$:

When $\theta = 1$, the production of new knowledge is proportional to the existing stock of knowledge. Then we can simplify eq (ii) and

③ to $g_A(t) = B \alpha_L^\theta L(t)^\theta$

$$g_A(t) = \theta n g_A(t)$$

If population growth, n , is positive, g_A increases overtime, since

If $n = 0$, g_A remains constant and the economy immediately exhibits steady growth.

(c) The expression of the $\dot{g}_k = 0$ and $\dot{g}_A = 0$ lines are given by

(1) $\dot{g}_k = 0 \Rightarrow g_k = g_A + n$

(2) $\dot{g}_A = 0 \Rightarrow g_k = \frac{(1-\theta)g_A - \theta n}{\beta}$

The expressions for the growth rates of capital and knowledge are:

(3) $g_k(t) = C_k [A(t)L(t)/K(t)]^{1-\alpha} n$

$$C_k = s(1-\alpha_k)^\alpha (1-\alpha_L)^{1-\alpha}$$

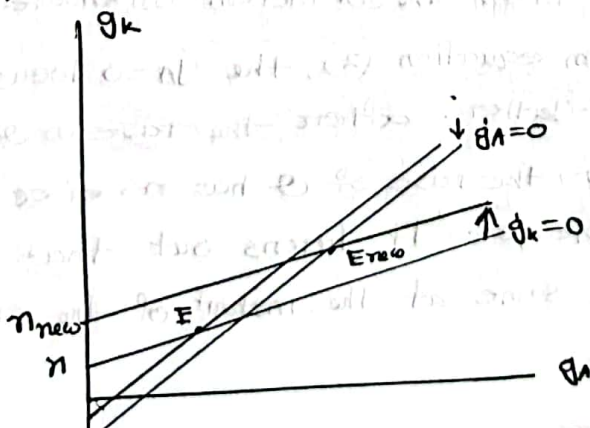
(4) $g_A(t) = C_A K(t)^\beta L(t)^\theta A(t)^{\theta-1}$

$$C_A = B \alpha_k^\beta \alpha_L^\theta$$

i) From eq (1), for a given g_A , the value of g_k satisfies $\dot{g}_k = 0$ is now higher as a result of the rise in population growth from n to n_{new} .

Thus $\dot{g}_k = 0$ locus shifts up. From eq (2), for a given g_A , the value of g_k that satisfies $\dot{g}_A = 0$ is now lower. Thus the $\dot{g}_A = 0$ locus shifts

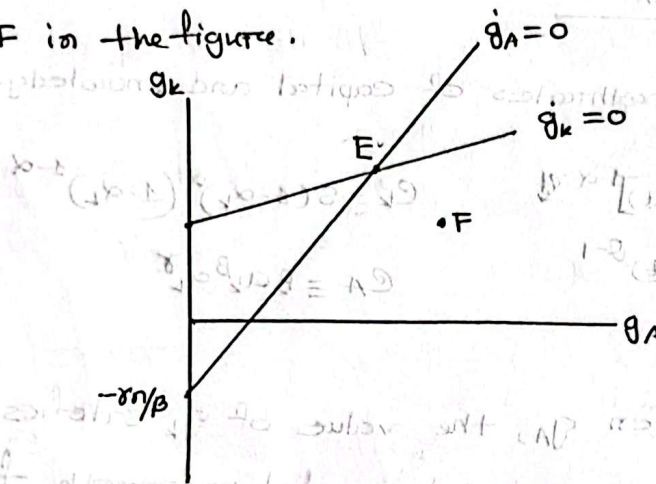
down.



Since n doesn't appear in eq(3), there is no jump in the value of g_k at the moment of the increase in population growth. Similarly, since n doesn't appear in eq(4), there is no jump in the value of g_A at the moment of rise in population growth.

ii) α_k doesn't appear in eq(1) and eq(ii). Thus, neither $\dot{g}_k=0$ nor $\dot{g}_A=0$ line shifts as a result of the increase in the fraction of the capital stock used in the knowledge sector.

From eq(3) the rise in α_k causes the growth rate of capital g_k to jump down. From eq(4) the growth rate of knowledge g_A jumps up at the instant of the rise in α_k . Thus the economy moves to a point such F in the figure.

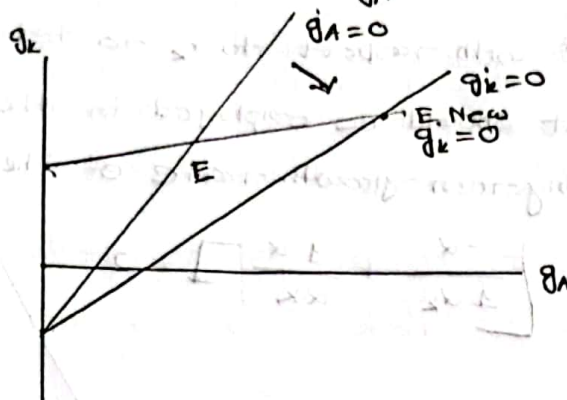


iii) Since θ doesn't appear in eq(1), there is no shift of the $\dot{g}_k=0$ locus as a result of the rise in θ , the coefficient on knowledge in the knowledge production function. From equation (2), the $\dot{g}_A=0$ locus has slope $(1-\theta)/\beta$ and therefore becomes flatter after the rise in θ . Since θ doesn't appear in eq(3) and (4), the rise of θ has no effect on the growth rate of capital and knowledge. It turns out that g_A may jump up, jump down or stay the same at the instant of the change in θ . From

eq (4) and (5) with respect we get,

$$\delta \ln g_A(t) / \delta \theta = \ln A(t)$$

So if $A(t)$ is less than 1, so that $\ln A(t) < 0$, the growth rate of knowledge jumps down at the instant of the rise in θ . However and vice versa. At $A(t) = 1$ at the time of the change in θ , there is no initial jump in g_A . This means the dynamics of adjustment to E_{new} may differ depending on the value of g_A at the time of the change in θ .



(d) i) The growth rates of capital and knowledge is:

$$g_k(t) \equiv \frac{\dot{k}(t)}{k(t)} = c_k [A(t)L(t)/k(t)]^{1-\alpha} \quad \text{where } c_k = s(1-\alpha)^\alpha (1-\alpha_L)^{1-\alpha}$$

$$g_A(t) = \frac{\dot{A}(t)}{A(t)} = c_A L(t)^\beta L(t)^\gamma A(t)^{\alpha-1} \quad \text{where } c_A = B\alpha_k^\beta \alpha_L^\gamma$$

With assumptions of $\beta + \gamma = 1$ and $n = 0$,

$$g_k(t) = [c_k L(t)^{1-\alpha}] [A(t)/k(t)]^{1-\alpha} \quad \text{and} \quad g_A(t) = [c_A L(t)^\beta] [k(t)/A(t)]^\beta$$

The two growth rates g_k and g_A will be equal when,

$$[c_k L(t)^{1-\alpha}] [A(t)/k(t)]^{1-\alpha} = [c_A L(t)^\beta] [k(t)/A(t)]^\beta$$

$$\Rightarrow [A(t)/k(t)]^{1-\alpha+\beta} = [c_A/c_k] L(t)^{\gamma-(1+\alpha)}$$

$$\Rightarrow A(t)/k(t) = \left[\left(\frac{c_A}{c_k} \right) L(t)^{\gamma-(1+\alpha)} \right]^{\frac{1}{1-\alpha+\beta}} \dots (ii)$$

ii) In order to find the growth rate of A and K when $g_K \equiv g_A = g^*$

substituting eq(ii) into (i)

$$g^* = [e_K L^{1-\alpha}] [(c_A/c_K) L^{\gamma-\beta-\alpha}]^{\frac{1-\alpha}{1-\alpha+\beta}} \\ = [e_K^{\frac{\alpha(1-\alpha+\beta)-(1-\alpha)}{1-\alpha+\beta}} c_A^{1-\alpha} L^{1-\alpha+\gamma(1-\alpha)-(1-\alpha)}]^{\frac{1}{1-\alpha+\beta}} \\ \therefore g^* = [e_K^\beta c_A^{1-\alpha} L^{(1-\alpha)(\gamma+\alpha)}]^{\frac{1}{1-\alpha+\beta}}$$

(iii) We can maximize $\ln g^*$ with respect to α_K to determine the fraction of the capital stock that should be employed in the R&D sector in order to maximize the long-run growth rate of the economy.

$$\frac{\partial \ln g^*}{\partial \alpha_K} = \frac{\beta}{(1-\alpha+\beta)} \left[\frac{-\alpha}{1-\alpha_K} + \frac{1-\alpha}{\alpha_K} \right] = 0$$

Solving for optimal

$$\frac{\alpha}{1-\alpha_K} = \frac{1-\alpha}{\alpha_K}$$

$$\Rightarrow \alpha \alpha_K = (1-\alpha)(1-\alpha_K)$$

$$\Rightarrow \alpha \alpha_K = 1 - \alpha_K - \alpha + \alpha \alpha_K$$

$$\Rightarrow \alpha_K^* = (1-\alpha)$$

Thus the optimal fraction for capital stock to employ in R&D sector is equal to effective labor's share in production of output. β , capital's share in the production function for new knowledge, does not affect the optimal allocation of capital to the R&D sector. The reason for this is that the increase in β has two effects. It makes capital more important in the R&D sector, thereby tending to raise the α_K that maximizes g^* . A rise in β also makes the production of new more valuable and new capital is produced when there is more output to be saved and

(iii) In order to see the way in which an increase in s affects the long-run growth rate we find the elasticity of the long-run growth rate of the economy with respect to saving rate:

$$\frac{\Delta \ln g^*}{\Delta \ln s} = \beta / (1 - \alpha + \beta) > 0$$

Thus an increase in the saving rate increases the long-run growth rate of the economy. This is essentially because it increases the resources devoted to physical capital accumulation and in this model, we have constant returns to produced factors of production.

invested. This tends to lower the α_k that maximizes g^* since it implies that more resources should be devoted to the production of output rather than knowledge. In the case we are considering these two effects exactly cancel each other.

✓ What is semi endogenous growth model? What is fully endogenous growth model?

The models in which long-run growth rate is endogenous and increases with population growth but the fraction of labor force and capital stock α_L and α_K do not affect the long-run growth, nor does the saving rate is called semi endogenous growth model. The model in which $\beta + \theta < 1$ or without capital $\theta < 1$ are semi endogenous growth model.

There ^{are} models in which increases in the saving rate and population size raise the long-run growth rate, and changes in α_L and α_K shift resources between good production and R&D, leading to ambiguous effects on long-run growth. Since long-run growth depends on various parameters, models where, $\beta + \theta > 1$ or $\beta + \theta = 1$ and $n > 0$ and $\theta > 1$ (without capital) are fully exogenous growth model.